

JEE MAIN REVISION MASTER: SOLID STATE OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 67 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Solid State - Core Application	67 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Solid State - Core Application (Very High Yield — 67 PYQs)

Core Principle: Key problem pattern in Solid State.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Which of the following expressions is correct in case of a CsCl unit cell (edge length 'a')?..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "A cubic solid is made up of two elements X and Y. Atoms of X are present on every alternate corner and one at the center of cube. Y is at \$\\dots\$." [Ans: N/A]
- [2023 Evening Shift] Q#1 (Numerical): "Number of crystal systems from the following where body centred unit cell can be found, is _____. Cubic, tetragonal, orthorhombic, he..." [Ans: 3]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

Which of the following expressions is correct in case of a CsCl unit cell (edge length 'a')?

- (A) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{\sqrt{3}}{2}a$
- (B) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{a}{\sqrt{2}}$
- (C) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = a$
- (D) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = (a / 2)$

Correct Answer: None

Solution Analysis: CsCl has body centered type structure in which Cs^+ occupies at corner of a cube and Cl^- occupies the centre of the cube. $2r_{\text{Cs}^+} + 2r_{\text{Cl}^-} = \sqrt{3}a$ (where a is the edge length of the cube) $r_{\text{Cs}^+} + r_{\text{Cl}^-} = \frac{\sqrt{3}}{2}a$